

AI CUSTOMER SUPPORT KNOWLEDGE BASE

Version 1.0 | Synthetic knowledge document for Retrieval-Augmented Generation (RAG)

Purpose: This document is designed to be ingested into a vector database such as ChromaDB. It provides clear policies, troubleshooting guidance, response boundaries, and escalation rules for an AI-powered customer support system.

1. Company and Product Overview

AcmeCloud is a fictional SaaS platform that provides cloud-based project management, team collaboration, file sharing, and workflow automation. Customers can access AcmeCloud through a web application and mobile applications.

Support hours are Monday through Friday, 09:00 to 18:00 UTC. Automated support is available at all times, but requests requiring human review are handled during support hours.

2. Account Access and Authentication

Password Reset

Customers can reset their password from the Sign In page by selecting 'Forgot Password'. A password reset link is sent to the registered email address. Reset links expire after 30 minutes. Customers should check their spam folder if the email is not received within 10 minutes.

No Access to Registered Email

If a customer no longer has access to their registered email address, automated support must not change the account email address. The case must be escalated for identity verification.

Locked Accounts

Accounts may be temporarily locked after five unsuccessful login attempts within 15 minutes. The lock automatically expires after 30 minutes. Support cannot manually bypass security locks without verification.

Two-Factor Authentication

Customers can enable two-factor authentication in Account Settings. If a customer loses access to their authentication device, the issue requires identity verification and human review. Automated support must not disable two-factor authentication.

3. Billing and Subscriptions

Available Plans

AcmeCloud offers Free, Pro, and Business plans. The Free plan has basic collaboration features. Pro includes advanced automation and increased storage. Business includes administrative controls, audit logs, and priority support.

Subscription Changes

Customers can upgrade or downgrade from the Billing section of their account. Upgrades take effect immediately. Downgrades take effect at the end of the current billing period unless otherwise stated during checkout.

Failed Payments

If a payment fails, the customer should verify card details, available funds, expiration date, and any bank restrictions on online or international transactions. After updating payment information, the system automatically retries the payment.

Duplicate Charges

If a customer believes they were charged twice, automated support should collect the transaction date, amount, and last four digits of the payment method if voluntarily provided. The case must then be escalated for billing investigation.

Invoice Requests

Customers can download available invoices from Billing > Payment History. If an invoice contains incorrect company information, the customer should submit the requested correction details for human review.

4. Refund Policy

Monthly subscriptions may be eligible for a refund if the request is submitted within 14 days of the original purchase and the customer has not substantially consumed premium services.

Annual subscriptions may be eligible for a prorated refund only when required by applicable law or approved as an exception by the billing team.

Refund eligibility is determined by the billing system and support policy. Automated support must not promise that a refund will be approved. It may explain the policy and submit or escalate a refund request.

Confirmed duplicate charges and unauthorized transactions always require human investigation before a financial outcome is communicated.

5. Technical Troubleshooting

Application Will Not Load

Ask the customer to refresh the page, sign out and sign in again, clear browser cache, disable conflicting browser extensions, and test using an updated supported browser. If the issue continues, collect the browser name, version, approximate time of the error, and any visible error message.

File Upload Failures

Verify that the file size is within the account limit, the file format is supported, and the internet connection is stable. Customers should retry using a different browser or network when possible.

Synchronization Problems

Ask the customer to verify their internet connection, refresh the application, and check whether the issue occurs on another device. Persistent synchronization problems should be escalated with timestamps and affected project or workspace identifiers.

Known Service Outages

When an official incident is active, support should explain that the engineering team is investigating, provide the latest known status, and avoid estimating a resolution time unless an official estimate is available.

6. Data, Privacy, and Security

Data Export

Customers may export eligible account data from Settings > Data Management. Large exports can take up to 24 hours to prepare.

Account Deletion

Account deletion requests require confirmation because deletion may permanently remove data. Automated support may explain the process but should not delete an account without the platform's authenticated confirmation workflow.

Personal Data Requests

Requests involving access to, correction of, or deletion of personal data may require identity verification. Cases involving legal rights or regulatory requests must be escalated.

Security Incidents

Suspected account compromise, unauthorized access, leaked credentials, suspicious API activity, or possible data exposure must be immediately escalated. Automated support should advise the customer to change their password and secure active sessions when those actions are available, but must not investigate or speculate about the cause.

7. Feature Requests and Product Feedback

Feature requests should be categorized by product area and summarized clearly. Automated support may acknowledge the request but must not promise that a feature will be implemented or provide an unconfirmed release date.

For feedback about an existing feature, the agent should determine whether the issue is a usage question, a bug, or a feature request before selecting the next action.

8. Escalation Rules

The AI agent must escalate rather than provide a final autonomous resolution when any of the following conditions apply:

1. The agent's confidence in the answer is below the configured confidence threshold.
2. The knowledge base does not contain sufficient information to answer safely.
3. The customer reports unauthorized access, suspected fraud, or a security incident.
4. The request requires identity verification or changes to account ownership or authentication.
5. The request involves a disputed charge, duplicate charge, or refund decision.
6. The customer reports potential data loss or data exposure.
7. The issue requires internal system access unavailable to the automated agent.
8. The customer explicitly requests a human support representative.
9. The request involves legal, regulatory, or privacy-rights questions.
10. The AI detects contradictory information or cannot reliably classify the ticket.

9. Response Policy for the AI Agent

The AI agent should answer only when the retrieved knowledge directly supports the response. It must not invent policies, account status, refund outcomes, incident causes, or product capabilities.

When confidence is high, the response should be concise, actionable, and include clear next steps.

When confidence is low or an escalation rule is triggered, the agent should explain that the request is being sent for human review and summarize the information already collected.

The agent should never claim to have performed an action unless the connected backend or tool confirms that the action was completed.

10. Recommended Ticket Categories

Category	Examples	Default Action
Account	Login, password reset, locked account	Answer or escalate
Billing	Invoices, payment failure, duplicate charge	Answer or escalate
Subscription	Upgrade, downgrade, cancellation	Answer when supported
Refund	Refund eligibility or request	Explain policy and review
Technical	Errors, uploads, sync issues	Troubleshoot or escalate
Security	Account compromise, suspicious activity	Immediate escalation
Privacy	Data export, deletion, legal requests	Answer or escalate
Feature Request	New functionality suggestions	Capture and acknowledge
Human Request	Customer asks for an agent	Pause for human review

11. Example Safe Responses

High Confidence Example

You can reset your password by selecting 'Forgot Password' on the Sign In page. The reset link expires after 30 minutes. If you do not see the email within 10 minutes, please check your spam folder.

Low Confidence Example

I do not have enough verified information to give you a reliable answer. I will send this case for human review so the support team can investigate it.

Security Escalation Example

Because this may involve unauthorized access to your account, this case needs immediate human review. Please change your password if you still have access to the account and avoid sharing passwords or authentication codes in this chat.

Refund Example

Refund eligibility depends on the subscription type, purchase date, and account usage. I can explain the policy, but I cannot guarantee approval. Your request can be reviewed by the billing team.

12. Knowledge Base Usage Notes

For best RAG performance, ingest this document using semantic chunking or a recursive text splitter. A chunk size of approximately 400 to 800 tokens with moderate overlap is a reasonable starting point. Preserve section headings in chunk metadata when possible.

Recommended metadata fields include: source, section, category, priority, escalation_required, and version. This enables the LangGraph agent to filter or prioritize retrieved documents before generating a response.

This document is synthetic and intentionally structured for an AI agent portfolio project. You can extend it later with FAQs, product documentation, incident reports, and policy updates to demonstrate a larger retrieval corpus.